

The Hong Kong University of Science and Technology
IEDA 1250, Spring 2026
Final Exam **Solution**

Student ID _____

Name: _____

Please read the following instructions carefully.

- You have **120** minutes to complete this exam. This question booklet contains **7** questions for the total of 120 points. Check to see if any pages are missing.
- Read the instructions for individual questions carefully before answering the questions.
- **Please write your answer in the answer book.** You can use this question booklet as scratch paper.

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
6	10	
7	10	
Total	120	

- (20 points) The Integrated Electronic Diagnostic Apparatus Company produces precision medical diagnostic equipment at two factories. Three medical centers have placed orders for this month's production output. The table below shows the unit shipping cost from each factory to each customer. It also shows the number of units produced at each factory and the number of units ordered by each customer.

From/To	Unit Shipping Cost			Output
	Customer 1	Customer 2	Customer 3	
Factory 1	\$600	\$800	\$700	400 units
Factory 2	\$400	\$900	\$600	500 units
Order size	300 units	200 units	400 units	

The company needs to decide how many units to ship from each factory to each customer to minimize total shipping cost. Each factory must ship exactly all units it produces, and each customer must receive exactly the number of units ordered.

- Formulate a linear programming (LP) model for this problem. Define your decision variables clearly.
- Derive the dual of the LP in Part 1.
- Suppose that an optimal primal shipping plan is given as follows:
 - Factory 1 ships 200 units to Customer 2 and 200 units to Customer 3.
 - Factory 2 ships 300 units to Customer 1 and 200 units to Customer 3.

Using complementary slackness, find one optimal dual solution. Because the optimal dual solution is not unique, you may set any one dual variable to zero and solve the others.

- If the dual LP is written correctly, each dual constraint should contain two dual variables, one associated with a factory and one associated with a customer. Suppose we interpret these two dual variables as prices charged by the transportation company to the factory and the customer, respectively. Explain the economic meaning of the dual constraint. At optimality, some dual constraints hold with equalities. What does this equality mean economically?

Solution

- Let x_{ij} be the number of units shipped from Factory i to Customer j , where $i = 1, 2$ and $j = 1, 2, 3$. The LP is

$$\begin{aligned}
 \min \quad & 600x_{11} + 800x_{12} + 700x_{13} + 400x_{21} + 900x_{22} + 600x_{23} \\
 \text{s.t.} \quad & x_{11} + x_{12} + x_{13} = 400, \\
 & x_{21} + x_{22} + x_{23} = 500, \\
 & x_{11} + x_{21} = 300,
 \end{aligned}$$

$$\begin{aligned}
x_{12} + x_{22} &= 200, \\
x_{13} + x_{23} &= 400, \\
x_{ij} &\geq 0, \quad i = 1, 2, \quad j = 1, 2, 3.
\end{aligned}$$

2. Let u_i be the dual variable associated with the supply constraint of Factory i , and let v_j be the dual variable associated with the demand constraint of Customer j . Since all primal constraints are equality constraints, all dual variables are unrestricted in sign. The dual is

$$\begin{aligned}
\max \quad & 400u_1 + 500u_2 + 300v_1 + 200v_2 + 400v_3 \\
\text{s.t.} \quad & u_1 + v_1 \leq 600, \\
& u_1 + v_2 \leq 800, \\
& u_1 + v_3 \leq 700, \\
& u_2 + v_1 \leq 400, \\
& u_2 + v_2 \leq 900, \\
& u_2 + v_3 \leq 600, \\
& u_1, u_2, v_1, v_2, v_3 \text{ unrestricted.}
\end{aligned}$$

3. The optimal shipments are

$$x_{12}^* = 200, \quad x_{13}^* = 200, \quad x_{21}^* = 300, \quad x_{23}^* = 200.$$

By complementary slackness, the corresponding dual constraints must be tight:

$$\begin{aligned}
u_1 + v_2 &= 800, \\
u_1 + v_3 &= 700, \\
u_2 + v_1 &= 400, \\
u_2 + v_3 &= 600.
\end{aligned}$$

Set $u_1 = 0$. Then

$$v_2 = 800, \quad v_3 = 700.$$

From $u_2 + v_3 = 600$, we get

$$u_2 = -100.$$

From $u_2 + v_1 = 400$, we get

$$v_1 = 500.$$

Therefore, one optimal dual solution is

$$u_1^* = 0, \quad u_2^* = -100, \quad v_1^* = 500, \quad v_2^* = 800, \quad v_3^* = 700.$$

The dual objective value is

$$400(0) + 500(-100) + 300(500) + 200(800) + 400(700) = 540000.$$

The cost of the given primal solution is

$$200(800) + 200(700) + 300(400) + 200(600) = 540000.$$

Thus the primal and dual objective values agree, so this is an optimal dual solution.

4. The dual constraint

$$u_i + v_j \leq c_{ij}$$

says that the total imputed price associated with Factory i and Customer j cannot exceed the unit shipping cost on route (i, j) . That is, the transportation company cannot charge the factory i and the customer j more than its cost, as such cost is known to everyone.

(Alternative: The dual constraint

$$u_i + v_j \leq c_{ij}$$

says that the combined shadow value assigned to Factory i and Customer j cannot exceed the unit shipping cost on route (i, j) . Otherwise, the dual prices would overvalue that route relative to its actual transportation cost.)

At optimality, if route (i, j) is used in the optimal shipping plan, then $x_{ij} > 0$, and complementary slackness implies

$$u_i + v_j = c_{ij}.$$

Economically, this means that for every route that is actually used, the unit shipping cost is exactly decomposed into the factory-side and customer-side shadow prices. If a route has strict inequality, then its reduced cost is positive, and that route is not attractive enough to be used in the optimal shipping plan.

2. (20 points) The Innovative Equipment Design and Automation Company has discontinued the production of a certain unprofitable product line. This act created considerable excess production capacity. Management is considering devoting this excess capacity to one or more of three products, call them products 1, 2, and 3. The available capacity on the machines that might limit output is summarized in the following table:

Machine Type	Available Time (Machine Hours per Week)
Milling machine	500
Lathe	350
Grinder	150

The number of machine hours required for each unit of the respective products is

Productivity coefficient (in machine hours per unit)			
Machine Type	Product 1	Product 2	Product 3
Milling machine	9	3	5
Lathe	5	4	0
Grinder	3	0	2

The sales department indicates that the sales potential for products 1 and 2 exceeds the maximum production rate and that the sales potential for product 3 is 20 units per week. The unit profit would be \$50, \$20, and \$25, respectively, on products 1, 2, and 3. The objective is to determine how much of each product the company should produce to maximize profit.

- Formulate the basic production planning problem as a linear program without the additional logical restrictions below. Explain the meaning of your decision variables.
- Management now has decided to add the restriction that
 - No more than two of the three prospective products should be produced.
 - If a product is produced, then at least 10 units of that product must be produced.
 - Product 3 can be produced only if product 2 is also produced.
 - Products 1 and 3 cannot be produced in the same week because they require incompatible tooling.
 - If product 1 is produced, then product 2 must also be produced.
 - If product 2 is the only product produced, then at least 60 units of product 2 must be produced.

Formulate these restrictions as logical constraints. Clearly explain any additional binary variables that you introduce.

Solution

1. Let x_i be the number of units of product i produced per week, for $i = 1, 2, 3$. The basic production planning LP is

$$\begin{aligned} \max \quad & 50x_1 + 20x_2 + 25x_3 \\ \text{s.t.} \quad & 9x_1 + 3x_2 + 5x_3 \leq 500, \\ & 5x_1 + 4x_2 \leq 350, \\ & 3x_1 + 2x_3 \leq 150, \\ & x_3 \leq 20, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Introduce binary variables

$$y_i = \begin{cases} 1, & \text{if product } i \text{ is produced,} \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2, 3.$$

The logical restrictions can be formulated as follows:

- No more than two of the three prospective products should be produced.

$$y_1 + y_2 + y_3 \leq 2$$

- If a product is produced, then at least 10 units of that product must be produced.

$$\begin{aligned} 10y_i &\leq x_i, \\ x_i &\leq y_i M, \end{aligned}$$

where $i = 1, 2, 3$, and M is a big positive number, such as valid upper bounds on the production quantity.

- Product 3 can be produced only if product 2 is also produced.

$$y_3 \leq y_2$$

- Products 1 and 3 cannot be produced in the same week because they require incompatible tooling.

$$y_1 + y_3 \leq 1$$

- If product 1 is produced, then product 2 must also be produced.

$$y_1 \leq y_2$$

- If product 2 is the only product produced, then at least 60 units of product 2 must be produced.

$$x_2 \geq 60(y_2 - y_1 - y_3),$$

3. (20 points) Consider the following zero-sum two-player game. The entries in the table are the payoffs to player A.

	B1	B2	B3	B4	B5	B6
A1	1	10	2	3	11	12
A2	4	8	5	6	9	10
A3	7	5	8	9	6	7
A4	10	1	11	12	2	3
A5	3	7	4	5	8	9
A6	6	4	7	8	5	6

Player A chooses one of the six rows, and player B chooses one of the six columns. Player A wants to maximize the expected payoff, while player B wants to minimize the expected payoff.

- Find the Nash equilibrium in this game and report the value of the game.
- Write player A's linear program for the *full* payoff table.
- The following linear program is associated with player B. Let q_j be the probability that player B chooses column j , for $j = 1, \dots, 6$, and let v_B represent the expected loss of player B:

$$\begin{aligned}
 \min \quad & v_B \\
 \text{s.t.} \quad & q_1 + 10q_2 + 2q_3 + 3q_4 + 11q_5 + 12q_6 \leq v_B, \\
 & 4q_1 + 8q_2 + 5q_3 + 6q_4 + 9q_5 + 10q_6 \leq v_B, \\
 & 7q_1 + 5q_2 + 8q_3 + 9q_4 + 6q_5 + 7q_6 \leq v_B, \\
 & 10q_1 + q_2 + 11q_3 + 12q_4 + 2q_5 + 3q_6 \leq v_B, \\
 & 3q_1 + 7q_2 + 4q_3 + 5q_4 + 8q_5 + 9q_6 \leq v_B, \\
 & 6q_1 + 4q_2 + 7q_3 + 8q_4 + 5q_5 + 6q_6 \leq v_B, \\
 & q_1 + q_2 + q_3 + q_4 + q_5 + q_6 = 1, \\
 & q_1, q_2, q_3, q_4, q_5, q_6 \geq 0.
 \end{aligned}$$

Show that player A's LP and player B's LP are primal and dual LPs. That is, starting from player A's LP in Part (ii), you need to derive its dual and show that the dual is identical to the given one.

Solution.

- First eliminate dominated strategies. For player A, row A5 is strictly dominated by row A2, and row A6 is strictly dominated by row A3. For player B, columns B3 and B4 are dominated by B1, and columns B5 and B6 are dominated by B2. The

reduced payoff matrix is

	<i>B1</i>	<i>B2</i>
<i>A1</i>	1	10
<i>A2</i>	4	8
<i>A3</i>	7	5
<i>A4</i>	10	1

Let q be the probability that player B chooses *B1*. Then player B chooses *B2* with probability $1 - q$. For each pure strategy of player A, the expected payoff to player A is

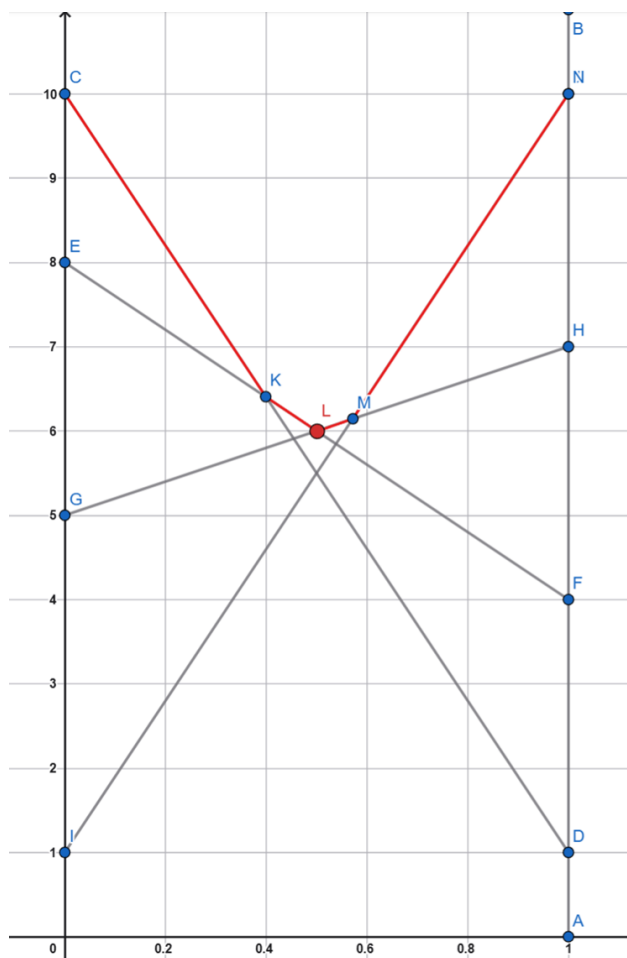
$$g_1(q) = q + (1 - q)10 = 10 - 9q,$$

$$g_2(q) = 4q + 8(1 - q) = 8 - 4q,$$

$$g_3(q) = 7q + 5(1 - q) = 5 + 2q,$$

$$g_4(q) = 10q + (1 - q) = 1 + 9q.$$

Graphically, we have



The relevant intersection is between $g_2(q)$ and $g_3(q)$:

$$8 - 4q = 5 + 2q.$$

Thus,

$$q = \frac{1}{2}.$$

Therefore, player B's optimal strategy is

$$q_1 = \frac{1}{2}, \quad q_2 = \frac{1}{2}, \quad q_3 = q_4 = q_5 = q_6 = 0.$$

The corresponding payoff for player A is 6, which is also the value of the game, i.e., $v = 6$.

It remains to find player A's optimal mixed strategy. Since $B1$ and $B2$ are both used by player B, player A should make player B indifferent between these two columns. Let player A choose $A2$ with probability p and $A3$ with probability $1 - p$. Then the expected payoff to player A if player B chooses $B1$ is

$$4p + 7(1 - p) = 7 - 3p,$$

and the expected payoff to player A if player B chooses $B2$ is

$$8p + 5(1 - p) = 5 + 3p.$$

Equating these two expressions gives

$$7 - 3p = 5 + 3p,$$

so

$$p = \frac{1}{3}.$$

Therefore, player A's optimal strategy is

$$p_1 = 0, \quad p_2 = \frac{1}{3}, \quad p_3 = \frac{2}{3}, \quad p_4 = p_5 = p_6 = 0.$$

Thus, the Nash equilibrium is

$$p^* = \left(0, \frac{1}{3}, \frac{2}{3}, 0, 0, 0\right), \quad q^* = \left(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right),$$

and the value of the game is

$$v = 6.$$

2. Let p_i be the probability that player A chooses row i , for $i = 1, \dots, 6$. Let v_A be the guaranteed expected payoff of player A. Player A's LP for the full payoff table

is

$$\begin{aligned}
\max \quad & v_A \\
\text{s.t.} \quad & p_1 + 4p_2 + 7p_3 + 10p_4 + 3p_5 + 6p_6 \geq v_A, \\
& 10p_1 + 8p_2 + 5p_3 + p_4 + 7p_5 + 4p_6 \geq v_A, \\
& 2p_1 + 5p_2 + 8p_3 + 11p_4 + 4p_5 + 7p_6 \geq v_A, \\
& 3p_1 + 6p_2 + 9p_3 + 12p_4 + 5p_5 + 8p_6 \geq v_A, \\
& 11p_1 + 9p_2 + 6p_3 + 2p_4 + 8p_5 + 5p_6 \geq v_A, \\
& 12p_1 + 10p_2 + 7p_3 + 3p_4 + 9p_5 + 6p_6 \geq v_A, \\
& p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1, \\
& p_1, p_2, p_3, p_4, p_5, p_6 \geq 0.
\end{aligned}$$

3. Students need to show the simplification process to obtain full score. There are two ways of doing the simplification. One is to rewrite the payoff constraints as

$$v_A - \sum_{i=1}^6 a_{ij}p_i \leq 0, \quad j = 1, \dots, 6.$$

and then write the dual LP.

However, if the students directly use

$$\sum_{i=1}^6 a_{ij}p_i - v_A \geq 0, \quad j = 1, \dots, 6,$$

then he/she has to change the sign of the corresponding dual variables and constraints.

4. (20 points) A college student has 7 days remaining before final examinations begin in her four courses, and she wants to allocate this study time as effectively as possible. She needs at least 1 day on each course, and she likes to concentrate on just one course each day, so she wants to allocate 1, 2, 3, or 4 days to each course. Having recently taken an optimization course, she decides to use dynamic programming to make these allocations to maximize the total grade points to be obtained from the four courses. She estimates that the alternative allocations for each course would yield the number of grade points shown in the following table:

Study Days	Course 1	Course 2	Course 3	Course 4
1	3	5	2	6
2	5	5	4	7
3	6	6	7	9
4	7	9	8	9

Solve this program by dynamic programming (Just show the calculation in the backward induction procedure. You don't need to write down recursive equations).

Solution.

Let $F_i(s)$ denote the maximum total grade points obtainable from Courses $i, i+1, \dots, 4$ when s study days remain for these courses. Then the terminal condition is $F_5(\cdot) \equiv 0$.

For Course 4,

s	1	2	3	4
$F_4(s)$	6	7	9	9

For Course 3,

s	2	3	4	5
$F_3(s)$	8	10	13	14

The calculations are

$$F_3(2) = \max\{2 + 6\} = 8,$$

$$F_3(3) = \max\{2 + 7, 4 + 6\} = \max\{9, 10\} = 10,$$

$$F_3(4) = \max\{2 + 9, 4 + 7, 7 + 6\} = \max\{11, 11, 13\} = 13,$$

$$F_3(5) = \max\{2 + 9, 4 + 9, 7 + 7, 8 + 6\} = \max\{11, 13, 14, 14\} = 14.$$

For Course 2,

s	3	4	5	6
$F_2(s)$	13	15	18	19

The calculations are

$$\begin{aligned}F_2(3) &= \max\{5 + F_3(2)\} = 5 + 8 = 13, \\F_2(4) &= \max\{5 + F_3(3), 5 + F_3(2)\} \\&= \max\{5 + 10, 5 + 8\} = \max\{15, 13\} = 15, \\F_2(5) &= \max\{5 + F_3(4), 5 + F_3(3), 6 + F_3(2)\} \\&= \max\{5 + 13, 5 + 10, 6 + 8\} = \max\{18, 15, 14\} = 18, \\F_2(6) &= \max\{5 + F_3(5), 5 + F_3(4), 6 + F_3(3), 9 + F_3(2)\} \\&= \max\{5 + 14, 5 + 13, 6 + 10, 9 + 8\} \\&= \max\{19, 18, 16, 17\} = 19.\end{aligned}$$

Finally, for all four courses with 7 days,

$$\begin{aligned}F_1(7) &= \max\{3 + F_2(6), 5 + F_2(5), 6 + F_2(4), 7 + F_2(3)\} \\&= \max\{3 + 19, 5 + 18, 6 + 15, 7 + 13\} \\&= \max\{22, 23, 21, 20\} \\&= 23.\end{aligned}$$

The maximum is attained by assigning 2 days to Course 1. Then $F_2(5) = 18$ is attained by assigning 1 day to Course 2. Then $F_3(4) = 13$ is attained by assigning 3 days to Course 3. Finally, 1 day is assigned to Course 4.

Therefore, the optimal allocation is

$$(x_1, x_2, x_3, x_4) = (2, 1, 3, 1),$$

and the maximum total grade points are

$$5 + 5 + 7 + 6 = 23.$$

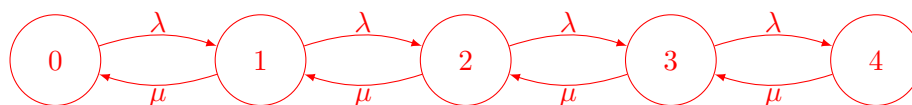
5. (20 points) Your system consists of a single CPU with finite buffer capacity. Jobs arrive according to a Poisson process with rate λ jobs/sec. The job sizes are Exponentially distributed with mean $1/\mu$ seconds. Jobs are serviced in FCFS order. Let $N - 1$ denote the maximum number of jobs that your system can hold in the queue. Thus, including the job serving, there are a maximum of N jobs in the system at any one time. If a job arrives when there are already N jobs in the system, then the arriving job is rejected.

Your Defense Advanced Research Projects Agency (DARPA) proposal requires that you reduce the probability of rejecting jobs in your system. To do this you could either ask for money to double the system capacity, or, alternatively, you could ask for money to double the speed of the CPU so that jobs get processed faster, thereby lowering the probability that there are N jobs in the system. Assuming both proposals have the same cost, which do you choose? (Asking for both is too expensive.)

Suppose that $N = 4$, $\lambda = 2$, and $\mu = 4$, then these are the specific questions you should answer:

- Draw the CTMC.
- Derive the stationary probabilities π_n for $n = 0, \dots, 4$.
- What is the utilization of the system?
- What is the fraction of jobs turned away (loss probability)? Use the word PASTA in your explanation.
- What is $\mathbb{E}[N]$?
- What is $\mathbb{E}[T]$ for only those jobs that enter the system. Hint: Here you need to use the effective arrival rate $\lambda_e := \sum_{n=0}^N \lambda_n \pi_n$.
- Which would have the greater effect on lowering loss probability: doubling the size of N or doubling the CPU speed?
- Answer the same question as (g) except now $N = 4$, $\lambda = 3$, and $\mu = 4$.
- Explain intuitively why (g) and (h) resulted in different answers

Solution. (a) The CTMC is



(b) The balance equations for the CTMC are

$$\begin{aligned}
 \text{State 0 : } & \mu\pi_1 = \lambda\pi_0, \\
 \text{State 1 : } & \lambda\pi_0 + \mu\pi_2 = (\lambda + \mu)\pi_1, \\
 \text{State 2 : } & \lambda\pi_1 + \mu\pi_3 = (\lambda + \mu)\pi_2, \\
 \text{State 3 : } & \lambda\pi_2 + \mu\pi_4 = (\lambda + \mu)\pi_3, \\
 \text{State 4 : } & \lambda\pi_3 = \mu\pi_4.
 \end{aligned}$$

Define $\rho := \frac{\lambda}{\mu} = 0.5$. Solving the above balance equations gives

$$\pi_n = \rho^n \pi_0, \quad \text{for } n = 0, 1, \dots, 4.$$

Using the normalization condition, we can obtain the stationary probabilities as follows

$$\begin{aligned}\pi_0 &= \frac{16}{31} \approx 0.5161, \\ \pi_1 &= \frac{8}{31} \approx 0.2581, \\ \pi_2 &= \frac{4}{31} \approx 0.1290, \\ \pi_3 &= \frac{2}{31} \approx 0.0645, \\ \pi_4 &= \frac{1}{31} \approx 0.0323.\end{aligned}$$

(c) The utilization equals the probability that the system is busy, so it is

$$1 - \pi_0 = \frac{15}{31} \approx 0.4839$$

(d) For Poisson arrivals, PASTA (Poisson Arrivals See Time Averages) tells us that the fraction of arriving jobs that find the system full equals the steady-state probability of being full:

$$\mathbb{P}(\text{Loss}) = \pi_4 = \frac{1}{31} \approx 0.0323.$$

(e) By definition, the expected number of jobs in the system equals

$$\mathbb{E}[N] = \sum_{n=1}^4 n\pi_n = \frac{26}{31} \approx 0.8387$$

(f) The effective arrival rate is

$$\lambda_e = \lambda(1 - \pi_4) = 2 \left(1 - \frac{1}{31}\right) = \frac{60}{31} \approx 1.9355.$$

By Little's Law for admitted jobs,

$$\mathbb{E}[T] = \frac{\mathbb{E}[N]}{\lambda_e} = \frac{26/31}{60/31} = \frac{13}{30} \approx 0.4333.$$

(g) If we double the system capacity, then $N = 8$, while $\lambda = 2$ and $\mu = 4$, so $\rho = 1/2$. Using the recursion $\pi_n = \rho^n \pi_0$ with $n = 0, 1, \dots, 8$, we can compute

$$\pi_0 = \frac{1}{1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^8}} = \frac{256}{511} \approx 0.5010.$$

Hence, the new loss probability is

$$\pi_8 = \rho^8 \pi_0 = \frac{1}{2^8} \frac{256}{511} = \frac{1}{511} \approx 0.0020.$$

If instead we double the CPU speed, then $N = 4$, $\lambda = 2$, and $\mu = 8$, so $\rho = 1/4$. Using the recursion $\pi_n = \rho^n \pi_0$ with $n = 0, 1, \dots, 4$, we can compute

$$\pi_0 = \frac{1}{1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \frac{1}{4^4}} = \frac{256}{341} \approx 0.7507.$$

Hence, the new loss probability is

$$\pi_4 = \rho^4 \pi_0 = \frac{1}{4^4} \frac{256}{341} = \frac{1}{341} \approx 0.0029.$$

Since

$$\frac{1}{511} < \frac{1}{341},$$

doubling the system capacity has the greater effect on lowering the loss probability.

(h) Now $N = 4$, $\lambda = 3$, and $\mu = 4$, so

$$\rho = \frac{3}{4}.$$

If we double the system capacity, then $N = 8$, while $\lambda = 3$ and $\mu = 4$, so $\rho = 3/4$. Using the recursion $\pi_n = \rho^n \pi_0$ with $n = 0, 1, \dots, 8$, we can compute

$$\pi_0 = \frac{1}{1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \dots + \left(\frac{3}{4}\right)^8} = \frac{65536}{242461} \approx 0.2703.$$

Hence, the new loss probability is

$$\pi_8 = \rho^8 \pi_0 = \left(\frac{3}{4}\right)^8 \frac{65536}{242461} = \frac{6561}{242461} \approx 0.0271.$$

If instead we double the CPU speed, then $N = 4$, $\lambda = 3$, and $\mu = 8$, so $\rho = 3/8$. Using the recursion $\pi_n = \rho^n \pi_0$ with $n = 0, 1, \dots, 4$, we can compute

$$\pi_0 = \frac{1}{1 + \frac{3}{8} + \left(\frac{3}{8}\right)^2 + \left(\frac{3}{8}\right)^3 + \left(\frac{3}{8}\right)^4} = \frac{20480}{32525} \approx 0.6297.$$

Hence, the new loss probability is

$$\pi_4 = \rho^4 \pi_0 = \left(\frac{3}{8}\right)^4 \frac{20480}{32525} = \frac{81}{6505} \approx 0.0125.$$

Since

$$\frac{81}{6505} < \frac{6561}{242461},$$

doubling the CPU speed has the greater effect on lowering the loss probability.

When the system load is light ($\rho = 0.5$). The system seldom fills; four extra buffer slots almost eliminate the already-rare rejection. Doubling the service rate helps, but not as dramatically.

When the system load is heavy ($\rho = 0.75$). The server is busy most of the time; congestion, not space, is the real issue. Extra buffer stores jobs but many arrivals are still blocked. Speeding the CPU cuts congestion directly, giving a much larger reduction in loss probability.

6. (10 points) Suppose your computer center currently consists of a single server of speed μ . Jobs arrive according to a Poisson process with rate λ , and their service times are Exponentially distributed.

Suppose the current response time is considered intolerable by the users. A second, faster server, running at speed $\alpha\mu$ (for $\alpha > 1$), is purchased and added to the system. With the second server, the system becomes a variant of the M/M/2 queue where the service rates of the two processors are not identical. Denote the service rate of the first processor by $\mu_1 = \mu$ and the service rate of the second processor by $\mu_2 = \alpha\mu$, where $\mu_1 < \mu_2$. In such system, the rule is that when both servers are idle, the faster server is scheduled for service before the slower one. Define the utilization, ρ , for this system to be $\rho = \lambda/(\mu_1 + \mu_2)$.

- (a) Suppose you are told that the mean response time $\mathbb{E}[T]$ in the system is given by

$$\mathbb{E}[T] = \frac{1}{\lambda A(1 - \rho)^2},$$

where

$$A = \frac{\mu_1\mu_2(1 + 2\rho)}{\lambda(\lambda + \mu_2)} + \frac{1}{1 - \rho}.$$

Then determine the mean response number of jobs $\mathbb{E}[N]$ to the system (you do not need to simplify your answer).

Solution. By Little's Law $\mathbb{E}[N] = \lambda\mathbb{E}[T]$, we have

$$\mathbb{E}[N] = \lambda \cdot \frac{1}{\lambda A(1 - \rho)^2} = \frac{1}{A(1 - \rho)^2}.$$

7. (10 points) In Question 6, a hotshot consultant walks in and makes the radical proposal of disconnecting the original server entirely (i.e., simply letting the faster server run by itself). Clearly this makes sense with respect to power, but the consultant claims this is also a win for $\mathbb{E}[N]$. For a consulting fee of \$400/hr, what is the consultant thinking? Come up with an instance, with specific values for λ , μ_2 , and μ_1 for which the consultant is right. Also, explain intuitively what is happening. [If you find this problem interesting, you can think about a general criterion under which the consultant would be right... Throwing away servers is fun!]

Solution. Choose

$$\lambda = 1, \quad \mu_1 = 1, \quad \mu_2 = 3.$$

For the two-server system,

$$\rho = \frac{\lambda}{\mu_1 + \mu_2} = \frac{1}{4}.$$

Also,

$$\begin{aligned} A &= \frac{\mu_1 \mu_2 (1 + 2\rho)}{\lambda(\lambda + \mu_2)} + \frac{1}{1 - \rho} \\ &= \frac{1 \cdot 3 \left(1 + 2 \cdot \frac{1}{4}\right)}{1(1 + 3)} + \frac{1}{1 - \frac{1}{4}} \\ &= \frac{3 \cdot \frac{3}{2}}{4} + \frac{4}{3} \\ &= \frac{9}{8} + \frac{4}{3} \\ &= \frac{59}{24} \\ &\approx 2.4583. \end{aligned}$$

Therefore, by the formula from Question 6,

$$\mathbb{E}[N_{\text{two-server}}] = \frac{1}{A(1 - \rho)^2} = \frac{1}{\frac{59}{24} \left(\frac{3}{4}\right)^2} = \frac{128}{177} \approx 0.7231.$$

If the original slower server is disconnected, then the system becomes an M/M/1 queue with arrival rate $\lambda = 1$ and service rate $\mu_2 = 3$. The mean number of jobs in this M/M/1 queue is

$$\mathbb{E}[N_{\text{fast-only}}] = \frac{\lambda}{\mu_2 - \lambda} = \frac{1}{3 - 1} = \frac{1}{2} = 0.5.$$

Since

$$0.5 < \frac{128}{177} \approx 0.7231,$$

disconnecting the slower server reduces the mean number of jobs in the system for this instance.

Intuition: Under the “fast-server-first” discipline, an arrival that finds the fast server busy but the slower server empty is sent straight to the slow server. When traffic is

light ($\lambda \ll \mu_2$) the queue is rarely occupied, so this *occasionally unlucky routing* dominates the average delay: the job spends a whole $1/\mu_1$ time units in service instead of waiting a much shorter $1/\mu_2$ for the fast server to free up. If the speed ratio α is large enough and the load is well below μ_2 , keeping the slow server actually *hurts* performance.

A rough rule of thumb: keep the slow server only when λ is close to μ_2 (so that the fast server alone would be nearly saturated). When λ is a small fraction of μ_2 , “throwing away” the slow server can indeed be a win—justifying the consultant’s \$400 /hr fee!